

INFORMATION TECHNOLOGY-IT1						
CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
221EIT006	WEB TECHNOLOGIES AND WEB SECURITIES	PROGRAM ELECTIVE 2	3	0	0	3

Preamble: This course helps the learner to understand the fundamental ideas of Web Application Development using HTML5, CSS3, JavaScript and detailed knowledge about server-side programming using PHP and Frameworks.

Course Outcomes: After the completion of the course the student will be able to:

CO 1	Demonstrate an understanding in web application development using HTML and CSS3 (Cognitive Knowledge Level: Apply)
CO 2	Implement web applications using client-side scripting (Cognitive Knowledge Level: Apply)
CO 3	Implement web applications using server-side scripting (Cognitive Knowledge Level: Apply)
CO 4	Implement session management and object-oriented programming using PHP. (Cognitive Knowledge Level: Apply)
CO 5	Understand the principles of web security and encoding schemes (Cognitive Knowledge Level: Understand)

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	✓	✓	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓	✓	✓
CO 4	✓	✓	✓	✓	✓	✓	✓
CO 5	✓	✓	✓	✓	✓	✓	✓

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	20%
Apply	80%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation: 40 marks

Preparing a review article based on peer reviewed

Original publications (minimum 10 15 marks

Publications shall be referred):

Course based task/Seminar/Data 15 marks

Collection and interpretation:

Test paper, 1 no.: 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern: 60 marks

The end semester examination will be conducted by the respective College. There will be two parts; Part A and Part B. Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly. For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60$ %.

Syllabus**Module – 1 (8 hours)**

Introduction: Web essentials – Clients – Servers – Communications – Mark-up Languages – HTML 5 – XHTML – Style sheets – CSS3.

Module – 2 (8 hours)

Client-side Scripting: Client-side Scripting using JavaScript – Syntax – Control Structures – Arrays and Objects – In-built objects-Host Objects: Browsers and the DOM

Module: 3 (7 hours)

Server-side Scripting: Introduction to PHP – Variables – Control Structures – Syntax – Arrays – Built-in Arrays – Form Processing-Object Oriented PHP.

Module: 4 (8 hours)

Database Applications: PHP with MySQL – mysql interface – MySQLi interface – Object Oriented Version of MySQLi

Module: 5 (8 hours)

Introduction to security testing Fundamentals - HTTP – Client-side scripting - Server-side scripting, Basic observation - observing live request headers, observing live post data, highlighting and detecting JavaScript events. Web oriented data encoding – working with base-36, base-64, URL encoded and HTML entity data.



Model Question Paper

**A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY
M.TECH DEGREE EXAMINATION
.....SEMESTER**

Branch:

Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Five marks each)

1. List all of the generic Internet top-level domains
2. Differentiate between MySQL and MySQLi
3. List out the features of Security Testing
4. Explain the usage of switch case statement in PHP with example
5. Write JavaScript code that will create an Object with a property named color having a String value of red

PART B(Answer any five questions. Seven marks each)

6. Write a style sheet that will cause the li elements within any ol element to be numbered in an outline style: the top-level li elements should use uppercase Roman numerals, the next level uppercase letters, the next level lowercase Roman numerals, then lowercase letters, and finally decimal numerals at the fifth level
7. What is the purpose of the HTML title attribute (not element)? Describe a scenario in which this attribute might be particularly helpful to a Web site user
8. Assume that you are adding JavaScript code to an HTML document that contains a form with a submit button. You did not modify any of the HTML or add an event listener for the click event on this button, and yet the button stopped working after you added your code to the document. Give at least two possible explanations.
9. Contrast the IE6 and DOM approaches to modifying an intrinsic event attribute, giving at least one advantage of each approach over the other.
10. Write in detail about PHP sessions
11. Explain aggregate functions in MySQL with example
12. How Live Request Headers are observed with Firebug
13. Describe in a detailed way different types of Web oriented data encoding techniques

Syllabus and Course Plan

No	Topic	No. of Lectures
Module – 1 (8 hours)		
1.1	Introduction: Web essentials	2
1.2	Clients – Servers – Communications	2
1.3	Mark-up Languages – HTML 5 – XHTML	2
1.4	Style sheets – CSS3	2
Module – 2 (8 hours)		
2.1	Client Side Scripting using JavaScript – Syntax	2
2.2	Control Structures – Arrays and Objects	2
2.3	In-built objects	2
2.4	Host Objects: Browsers and the DOM	2
Module: 3 (8 hrs)		
3.1	Server-side Scripting: Introduction to PHP	2
3.2	Variables – Control Structures – Syntax	2
3.3	Arrays – Built-in Arrays – Form Processing	2
3.4	Object Oriented PHP	2
Module: 4 (8 hrs)		
4.1	Database Applications: PHP with MySQL	3
4.2	mysql interface	2
4.3	MySQLi interface	2
4.4	Object Oriented Version of MySQLi	1
Module: 5 (8 hrs)		
5.1	Introduction to security testing Fundamentals - HTML - HTTP	2
5.2	Client-side scripting - Server-side scripting	2
5.3	Basic observation observing live request headers, observing live post data, highlighting and detecting JavaScript events.	2
5.4	Web oriented data encoding – working with base36, base 64, URL encoded and HTML entity data.	2

Reference Books

1. Jeffrey C Jackson, —Web Technology – A Computer Science Perspective, Pearson Education, 2007.
2. Brian P Hogan, —HTML5 and CSS3: Develop with Tomorrow's Standards Today, Pragmatic Bookshelf, 2010.
3. W Jason Gilmore, —Beginning PHP and MySQL: From Novice to Professional, Apress, 2010.
4. Paco Hope, Ben Walther, —Web Security Testing Cookbook: Systematic Techniques to Find Problems Fast, O'REILLY media, 2009.

